

SOP:

In classical Cryptography we don't much more talk about key sharing mechanism rather talk about encryption and decryption mechanism. We don't restrict eavesdropping as Man In Middle Attack (to retrieve the encryption key) once revealed the whole security is compromised. So, Key exchanging between among Alice and Bob through "Trusted Repeater" is a challenging task any time the security can be compromised by different types of attack (Brute Force Attack, Template based attack, co-relation analysis, Frequency counting analysis, some time Side Channel Analysis (Differential Fault Attack, Cache Timing Attack etc.)

Drawback :

1. As Distance of Light Path increased signal become weak or noisy, so regenerate and amplify the signal before reach to a certain distance some time based on the physical property of "Fiber-Optic Cable". How to decide "Where to Setup a Trusted Repeater is Crucial. Hence the Optimization in terms of Distance measure to overcome the "Attenuation" & "Dispersion" Problem which must ensure "Low Latency/Delay" & with minimal noise(disrupted signal) such that OSNR>threshold such that Bit Error Rate/QBER in Quantum Communication should be minimum.
2. A "multi port" Repeater connects multiple wires coming from different branches(Hub) can't filter packets only transmit to their destination. Routing & wavelength Assignment is not efficient leads to wastage of data packets.

RWA:

goal:

1. Setting up a light path by RWA on a Set of connections.
2. if no wavelength is assigned on a light path then the connection request is **Blocked**.
3. Connection Request Type (Static, Incremental, Dynamic)

Minimizes: the amount of "Connection Blocking"

Maximizes: The no of connections that are established in the network at the same time.

"SLE" ---> Static Light Path Establishment

"DLE" ---> Dynamic Light Path Establishment"

Hence the **ILP (Integer Programming Problem)**

Routing -> Multi Community Flow formulation

Wavelength Assignment ---> Graph Coloring (NP-Complete)

Quantum Internet with Local Access (QuILA)

CHENNAI

Bengaluru

Delhi

550KM ,Fiber Optic

2200 KM ,

Fiber optic

NQM -> 6,003 Core Project

Why QKD?

In Classical Cryptography Key Sharing among one to one (Alice to Bob) or multiparty communication is very challenging to detect the presence of eavesdropper using classical properties based on Mathematical Hardness of factoring like RSA. But Shor algorithm does this in polynomial time. Hence Still insecure. So QKD(BB84) ensures the presence or absence of the eavesdropper in between of the communication with 100% guarantee based on the principal of quantum mechanics "Heisenberg Uncertainty" and "No-Cloning" properties.

BB84 is a “ **Prepare and measure**” algorithm . “**No-Cloning**” guarantee that nobody can copy an unknown quantum state ,Measurement by MIMT destroys that quantum state by some detectable way.

Some polarized photons are complementary to each other (Rectilinear Basis and Diagonal Basis), Measuring one property necessarily disturbs the others called “ **Heisenberg Uncertainty**”.

Why Trusted Repeater?

Entanglement Over the Long Distance can be achieved by “Trusted Repeater” by using the concept of “Entanglement Swapping”. Quantum Repeater creates long range Bell state (maximally entangled state) used in Quantum Communication such as “QKD” or “Teleportation”. Entanglement Distribution , Entanglement Purification and Entanglement Swapping are the three important properties related to it.

Also Important in **which Band** we are doing our transmission?

Fiber_optic--→low fiber transmission losses(0.17db/km),the Attenuation_length=25 km.

WDM -----→ **CWDM (C -band,S- band , L-band) & DWDM(L-band)**

L band has higher Attenuation and Chromatic Dispersion compered to **C band**. Also L band suffers from **NLI** (non-linear impairment). Some time uses C+L band.

Different Spectrum Slot Allocation Policy:

- 1. First-Fit**
- 2. Last Fit**
- 3.Random Fit**

In C+L band , “Last-Fit” starts filling C band channels , where Links (Attenuation , Amplifier Noise , NLI all are decreases) effect are lower compared to other end of the spectrum. Hence Last **Fit policy** leads to **higher OSNR** of light path so higher capacity of the network.